

1

Overview and Descriptive Statistics

1.3 Measures of Location

Measures of Location

More formal data analysis often requires the calculation and interpretation of numerical summary measures.

Measurements that characterize the data set and convey some of its salient features.

Focus on numerical data primarily, a little categorical data at the end.

Two important location measures are both for the center:

Mean and Median



The Mean

The Mean

For a given set of numbers $x_1, x_2, \dots, x_n$, the most familiar and useful measure of the center is the *mean*, or arithmetic average of the set.

The **sample mean** $\bar{x}$ of observations $x_1, x_2, \dots, x_n$, is

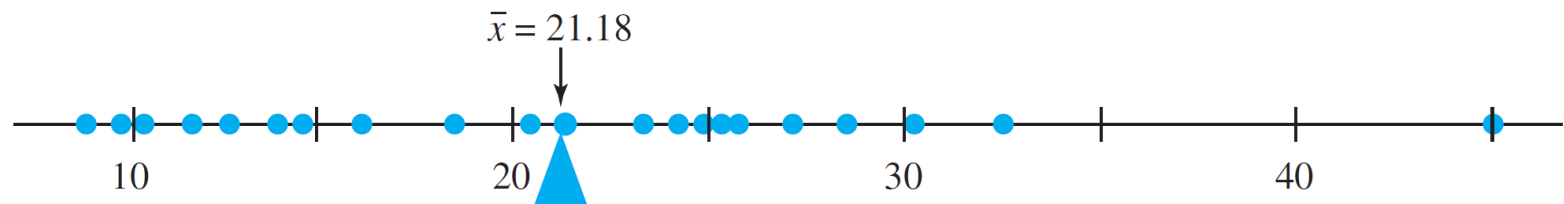
$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Recommend using decimal accuracy of one digit more than the accuracy of the x_i 's.

Example 14

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$
 $x_6 = 21.2$ $x_7 = 30.2$ $x_8 = 25.8$ $x_9 = 18.5$ $x_{10} = 10.3$
 $x_{11} = 25.3$ $x_{12} = 14.0$ $x_{13} = 27.1$ $x_{14} = 45.0$ $x_{15} = 23.3$
 $x_{16} = 24.2$ $x_{17} = 14.6$ $x_{18} = 8.9$ $x_{19} = 32.4$ $x_{20} = 11.8$
 $x_{21} = 28.5$

With $\sum x_i = 444.8$, the sample mean is $\bar{x} = \frac{444.8}{21} = 21.18$



The mean as the balance point for a system of weights

The Mean for population

The **population mean**, μ , is the average of all population values, $\mu = (\sum N \text{ values})/N$, for a finite population of size N .

Later for statistical inference, we will learn methods based on the sample mean to draw conclusions for a population mean.

For example, we can use the sample mean $\bar{x} = 21.18$ computed in Example 14 as a *point estimate* of $\mu =$ crack length for all specimens.

The Mean: one weakness

One deficiency to measure center: Its value can be greatly affected by an outlier (unusually large or small observation). In Example 14, $x_{14} = 45.0$.

Without x_{14} , $\bar{x} = 399.8/20 = 19.99$; the outlier increases the mean by more than $1 \mu\text{m}$. If the $45.0 \mu\text{m}$ observation were replaced by the catastrophic value $295.0 \mu\text{m}$ a really extreme outlier, then

$$\bar{x} = 694.8/21 = 33.09,$$

which is larger than all but one of the observations!

The Mean

A sample of incomes often produces such outlying values (those lucky few who earn astronomical amounts), and the use of average income is often misleading.

There is a measure that is less sensitive to outlying values than $\bar{x}$, the **median**.

However, the mean is still the most widely used measure, because there are many populations that don't have extreme outliers in the sample often.

Sampling from such a population (a normal or bell-shaped population), the sample mean will tend to be stable and quite representative of the sample.



The Median

The Median

The **sample median** $\tilde{x}$ is the middle value once the observations are ordered from smallest to largest, for $x_1 \leq x_2 \leq \dots, \leq x_n$.

$$\tilde{x} = \begin{cases} \text{The single middle value if } n \text{ is odd} & = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ ordered value} \\ \text{The average of the two middle values if } n \text{ is even} & = \text{average of } \left(\frac{n}{2} \right)^{\text{th}} \text{ and } \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ ordered values} \end{cases}$$

Example 15

cont'd

A sample of 12 recordings of Beethoven's Symphony #9, yielding the following durations (min) listed ascendingly:

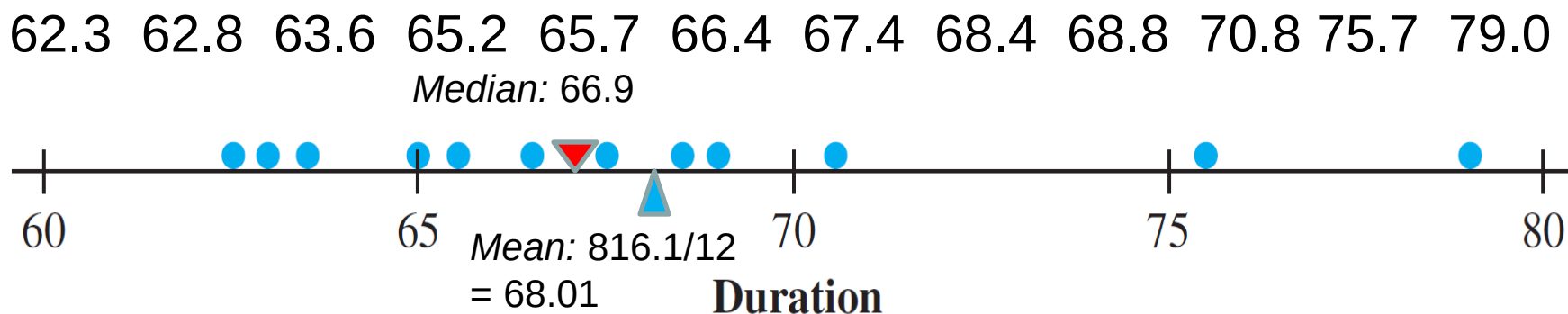


Figure 1.16 Dotplot of the data from Example 14

$n = 12$, the sample median is the average of the 6th and 7th values from the ordered list:

$$\tilde{x} = \frac{66.4 + 67.4}{2} = 66.90$$

Had the largest observation 79.0 not been there, the resulting sample median for the $n = 11$ remaining observations would have been the single middle (6th) value 66.4.

The Median

The data in Example 15 illustrates an important property of $\tilde{x}$ in contrast to $\bar{x}$: The sample median is insensitive to outliers. If, for example, we increased the two largest x_i s from 75.7 and 79.0 to 85.7 and 89.0, respectively, $\tilde{x}$ would be unaffected.

Thus, in the treatment of outlying data values, $\bar{x}$ and $\tilde{x}$ are at opposite ends of a spectrum. Both measures describe where the data is centered, but they will not in general be equal because they focus on different aspects of the sample.

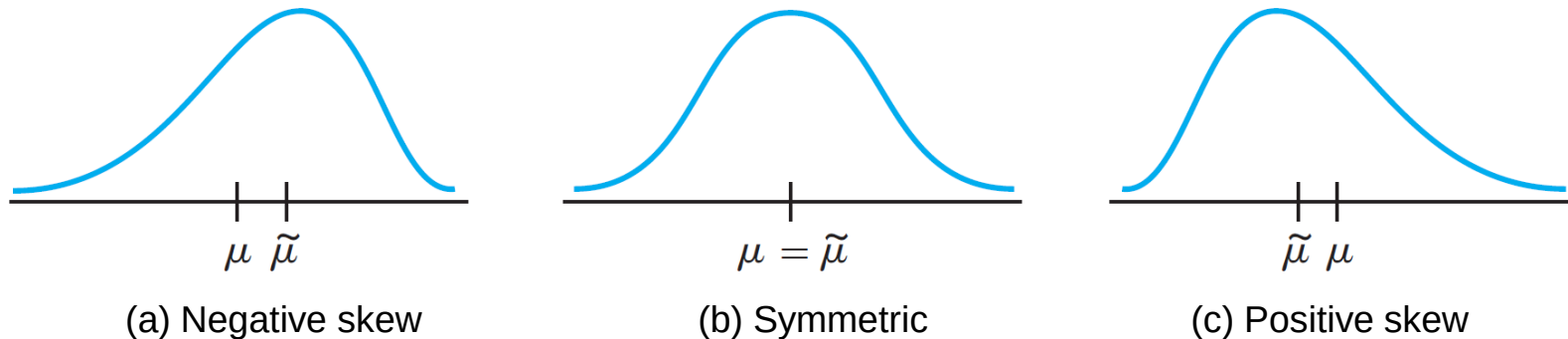
The Median

Analogous to $\tilde{x}$ as the middle value in the sample is a middle value in the population, the **population median**, denoted by $\tilde{\mu}$. As with $\bar{x}$ and μ , we can think of using the sample median $\tilde{x}$ to make an inference about $\tilde{\mu}$.

In Example 15, we might use $\bar{x} = 66.90$ as an estimate of the median time for the population of all recordings. A median is often used to describe income or salary data (because it is not greatly influenced by a few large salaries).

The Median

The population mean μ and $\tilde{\mu}$. median will not generally be identical. If the population distribution is positively or negatively skewed, as pictured in Figure 1.17, then $\mu \neq \tilde{\mu}$.



Three different shapes for a population distribution

Figure 1.17

When this is the case, in making inferences we must first decide which of the two population characteristics is of greater interest and then proceed accordingly.



Other Measures of Location: Quartiles, Percentiles, and Trimmed Means

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The median (population or sample) divides the data set into two parts of equal size. To obtain finer measures of location, we could divide the data into more than two such parts.

Roughly speaking, quartiles divide the data set into four equal parts, with the observations above the third quartile constituting the upper quarter of the data set, the second quartile being identical to the median, and the first quartile separating the lower quarter from the upper three-quarters.

Other Measures of Location: Quartiles, Percentiles, and Trimmed Means

Similarly, a data set (sample or population) can be even more finely divided using percentiles; the 99th percentile separates the highest 1% from the bottom 99%, and so on.

Unless the number of observations is a multiple of 100, care must be exercised in obtaining percentiles.

Other Measures of Location: Quartiles, Percentiles, and Trimmed Means

The mean is quite sensitive to a single outlier, whereas the median is impervious to many outliers. Since extreme behavior of either type might be undesirable, we briefly consider alternative measures that are neither as sensitive as $\bar{x}$ nor as insensitive as $\tilde{x}$.

To motivate these alternatives, note that $\bar{x}$ and $\tilde{x}$ are at opposite extremes of the same “family” of measures.

The mean is the average of all the data, whereas the median results from eliminating all but the middle one or two values and then averaging.

Other Measures of Location: Quartiles, Percentiles, and Trimmed Means

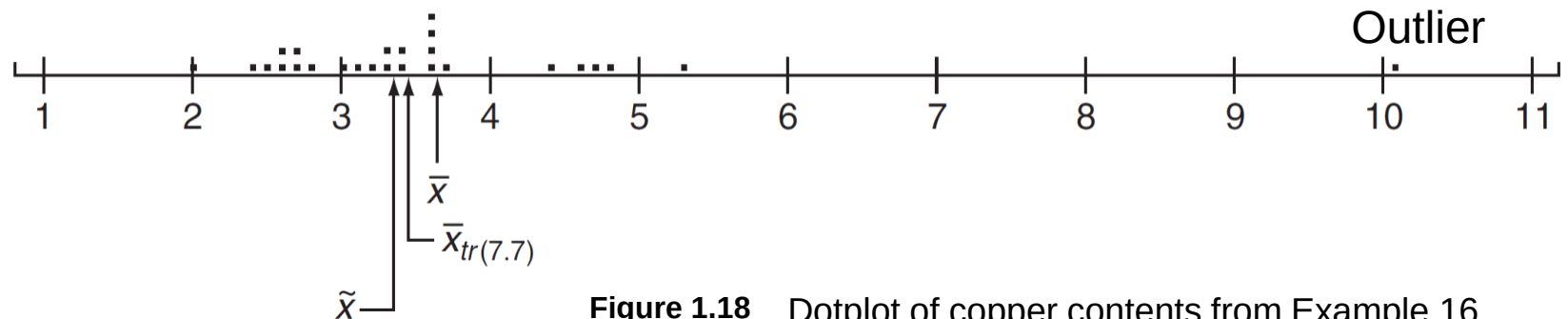
To paraphrase, the mean involves trimming 0% from each end of the sample, whereas for the median the maximum possible amount is trimmed from each end.

A **trimmed mean** is a compromise between $\bar{x}$ and $\tilde{x}$. A 10% trimmed mean, for example, would be computed by eliminating the smallest 10% and the largest 10% of the sample and then averaging what remains.

Example 16

Consider the following observations on copper content (%) for a sample of alloy:

2.0 2.4 2.5 2.6 2.6 2.7 2.7 2.8 3.0 3.1 3.2 3.3 3.3
3.4 3.4 3.6 3.6 3.6 3.6 3.7 4.4 4.6 4.7 4.8 5.3 10.1



The sample mean and median are 3.65 and 3.35, respectively. A trimmed mean with a trimming percentage of $100(2/26) = 7.7\%$ results from eliminating the two smallest and two largest observations; $\bar{x}_{tr(7.7)} = 3.42$.

Trimmed Mean: when desired % is not integer

If the desired trimming percentage is $100\alpha\%$ and $n\alpha$ is not an integer, the trimmed mean must be calculated by interpolation. For example, consider $\alpha = .10$ for a 10% trimming percentage and $n = 26$ as in Example 16.

Then $x_{tr(10)}$ would be the appropriate weighted average of the 7.7% trimmed mean calculated there and the 11.5% trimmed mean resulting from trimming three observations from each end.



Categorical Data and Sample Proportions

Categorical Data and Sample Proportions

When the data is categorical, the natural numerical summary quantities are the individual frequencies and the relative frequencies.

A **dichotomous** population—one having only two categories. If we let x denote the number in the sample in category 1 (the one we are more interested), then the number in category 2 is $n - x$. The relative frequencies or *sample proportions* in categories 1 and 2 are x/n and $1 - x/n$.

Categorical Data and Sample Proportions

Let's denote a response that falls in category 1 by a 1 and a response that falls in category 2 by a 0. A sample size of $n = 10$ might then yield the responses 1, 1, 0, 1, 1, 1, 0, 0, 1, 1. The sample mean for this numerical sample is:

$$\frac{x_1 + \cdots + x_n}{n} = \frac{1 + 1 + 0 + \cdots + 1 + 1}{10} = \frac{7}{10} = \frac{x}{n} = \text{sample proportion}$$

Then the sample proportion of observations in the category is the sample mean of the sequence of 1s and 0s. Thus a sample mean can be used to summarize the results of a categorical sample.

Categorical Data and Sample Proportions

Analogous to the sample proportion x/n of individuals or objects falling in a particular category, let p represent the proportion of the entire population falling in the category.

As with x/n , p is also between 0 and 1, and while x/n is a sample characteristic, p is a population characteristic.

The relationship between the two parallels the relationship between $\tilde{x}$ and $\tilde{\mu}$, and between $\bar{x}$ and μ . In particular, we will subsequently use x/n to make inferences about p .

Categorical Data and Sample Proportions

With k categories ($k > 2$), we can use the k sample proportions to answer questions about the population proportions $p_1, \dots, p_k$.